8. Write a program to implement Linear Regression (LR) algorithm in python

a)

CGPA	Aptitude Score	Communication Score	Placement
9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed
6.8	68	65	Not Placed
9	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed
6.5	60	58	Not Placed
8.9	88	86	Placed

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import LabelEncoder
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_1", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nActual Values:")
print(y_test)
print("\nPredicted Values:")
print(y_pred)
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
new_sample = [[8.9, 87, 84]]
```

```
prediction = model.predict(new_sample)
print("\nPredicted Value for New Sample:", prediction[0])
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Linear Regression - Placement Dataset")
plt.show()
```

OUTPUT:

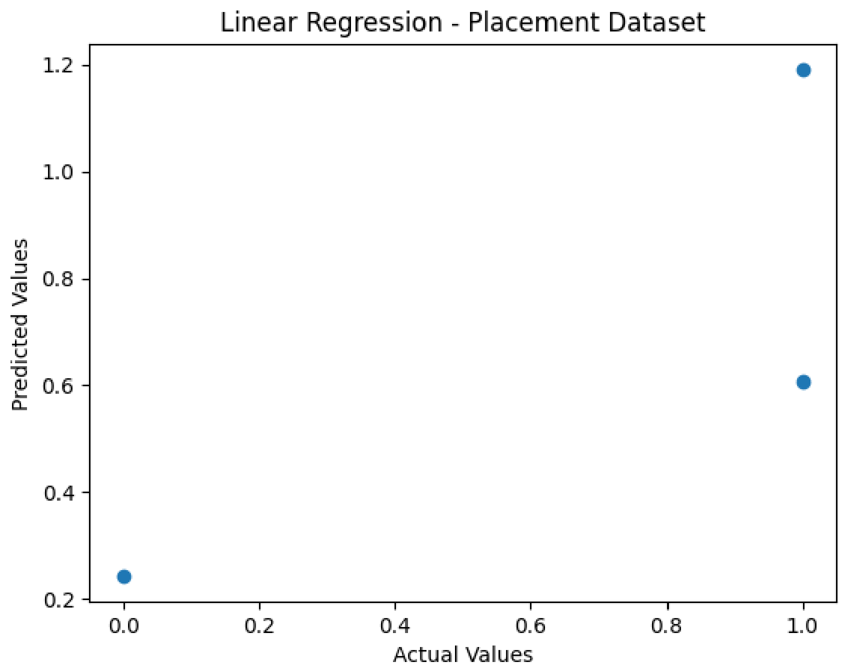
```
Dataset:
   CGPA  Aptitude Score  Communication Score   Placement
0   9.2              90                   88      Placed
1   8.8              85                   82      Placed
2   8.5              80                   79      Placed
3   7.2              72                   70  Not Placed
4   6.8              68                   65  Not Placed
5   9.0              92                   90      Placed
6   7.5              75                   74  Not Placed
7   8.6              84                   81      Placed
8   6.5              60                   58  Not Placed
9   8.9              88                   86      Placed

Actual Values:
[0 1 1]

Predicted Values:
[0.24372652 1.19053343 0.60601052]

Mean Squared Error: 0.08364443937672389
R2 Score: 0.6236000228047425

Predicted Value for New Sample: 1.12907588279489
```



b)

Income (Lakhs)	Credit Score	Loan Amount (Lakhs)	Loan Approved
8	780	5	Yes
7	760	4	Yes
6	720	6	Yes
4	620	8	No
3	580	10	No
9	800	4	Yes
5	650	7	No
8	770	5	Yes
4	600	9	No
7	740	6	Yes

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import LabelEncoder
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_2", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nActual Values:")
print(y_test)
print("\nPredicted Values:")
print(y_pred)
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
new_sample = [[7, 760, 5]]
prediction = model.predict(new_sample)
print("\nPredicted Value for New Sample:", prediction[0])
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
```

```
plt.ylabel("Predicted Values")
plt.title("Linear Regression - Loan Dataset")
plt.show()
```

OUTPUT:

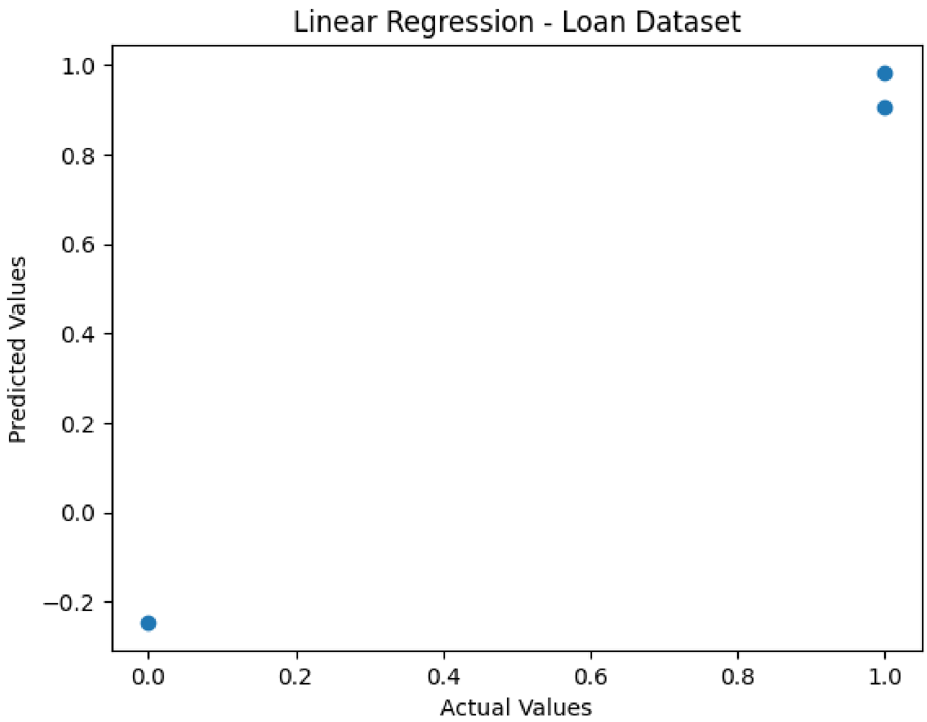
```
Dataset:
   Income (Lakhs)  Credit Score  Loan Amount (Lakhs) Loan Approved
0               8           780                    5           Yes
1               7           760                    4           Yes
2               6           720                    6           Yes
3               4           620                    8            No
4               3           580                   10            No
5               9           800                    4           Yes
6               5           650                    7            No
7               8           770                    5           Yes
8               4           600                    9            No
9               7           740                    6           Yes

Actual Values:
[0 1 1]

Predicted Values:
[-0.24675325  0.98376623  0.90584416]

Mean Squared Error: 0.023338674312700069
R2 Score: 0.8949759655928469

Predicted Value for New Sample: 1.194805194805193
```



c)

Temperature (°C)	Heart Rate (bpm)	Oxygen Level (%)	Disease
39	110	94	Positive
38.5	108	95	Positive
37	78	99	Negative
39.2	115	93	Positive
36.8	75	98	Negative
38.8	112	94	Positive
37.2	80	98	Negative
39.1	114	92	Positive
36.7	74	99	Negative
38.9	111	93	Positive

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import LabelEncoder
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_3", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nActual Values:")
print(y_test)
print("\nPredicted Values:")
print(y_pred)
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
new_sample = [[38.5, 108, 95]]
prediction = model.predict(new_sample)
print("\nPredicted Value for New Sample:", prediction[0])
plt.scatter(y_test, y_pred)
```

```
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Linear Regression - Disease Dataset")
plt.show()
```

OUTPUT:

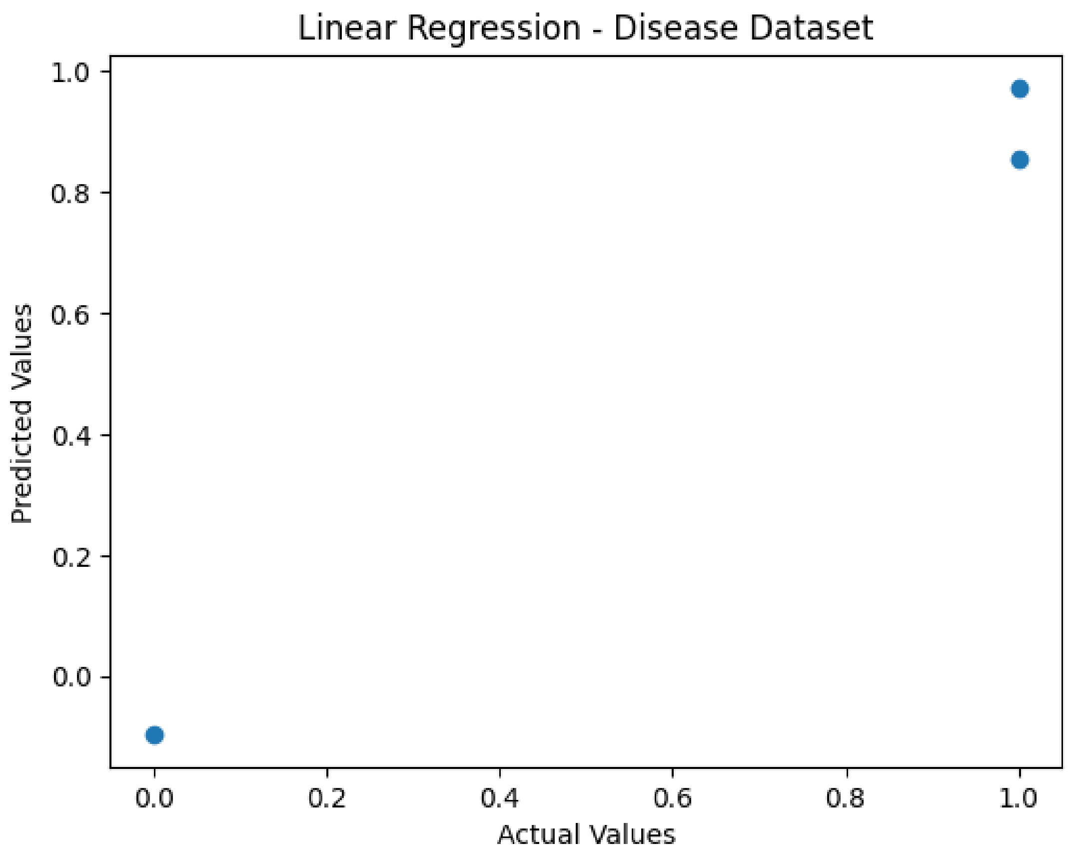
```
Dataset:
   Temperature (°C)  Heart Rate (bpm)  Oxygen Level (%)   Disease
0              39.0               110                94  Positive
1              38.5               108                95  Positive
2              37.0                78                99  Negative
3              39.2               115                93  Positive
4              36.8                75                98  Negative
5              38.8               112                94  Positive
6              37.2                80                98  Negative
7              39.1               114                92  Positive
8              36.7                74                99  Negative
9              38.9               111                93  Positive

Actual Values:
[0 1 1]

Predicted Values:
[-0.09601105  0.8561561   0.97271628]

Mean Squared Error: 0.010217863176833444
R2 Score: 0.9540196157042495

Predicted Value for New Sample: 0.8561561042997918
```



d)

Experience	Performance Score	Training Hours	Promotion
10	95	50	Yes
8	90	45	Yes
7	88	40	Yes
3	65	20	No
2	60	18	No
9	93	48	Yes
4	70	25	No
8	89	42	Yes
2	58	15	No
6	85	38	Yes

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import LabelEncoder
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_4", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("\nActual Values:")
print(y_test)
print("\nPredicted Values:")
print(y_pred)
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
new_sample = [[9, 92, 48]]
prediction = model.predict(new_sample)
print("\nPredicted Value for New Sample:", prediction[0])
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
```

```
plt.ylabel("Predicted Values")
plt.title("Linear Regression - Employee Dataset")
plt.show()
```

OUTPUT:

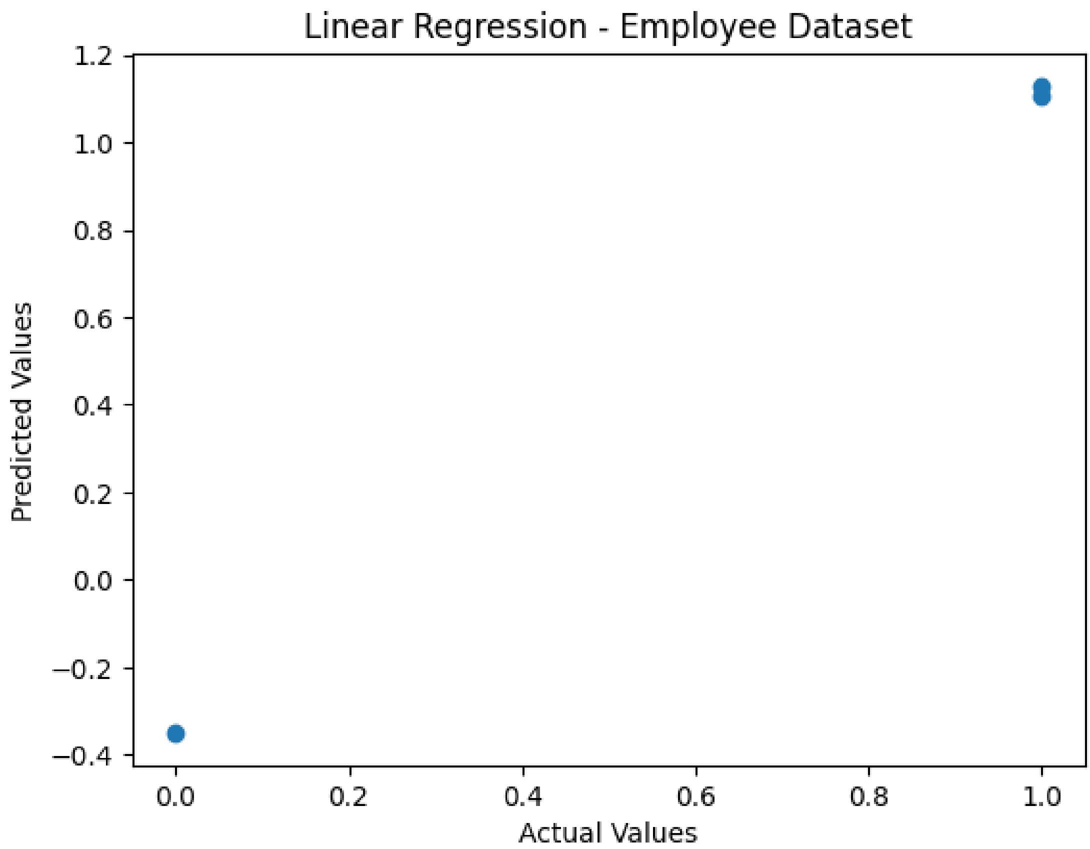
```
Dataset:
   Experience  Performance Score  Training Hours Promotion
0          10                 95              50       Yes
1           8                 90              45       Yes
2           7                 88              40       Yes
3           3                 65              20        No
4           2                 60              18        No
5           9                 93              48       Yes
6           4                 70              25        No
7           8                 89              42       Yes
8           2                 58              15        No
9           6                 85              38       Yes

Actual Values:
[0 1 1]

Predicted Values:
[-0.3499194   1.10478238  1.12831811]

Mean Squared Error: 0.049962822774248496
R2 Score: 0.7751672975158818

Predicted Value for New Sample: 1.0905964535196127
```



e)

Weight	Diameter	Sweetness	Fruit
150	7.5	90	Apple
160	7.8	88	Apple
140	7.2	91	Apple
120	5.5	70	Orange
125	5.8	72	Orange
130	6	74	Orange
155	7.6	89	Apple
118	5.4	69	Orange
148	7.4	92	Apple
122	5.7	71	Orange

PROGRAM:

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

from sklearn.preprocessing import LabelEncoder

df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_5", nrows=10)

print("Dataset:")

print(df)

X = df.iloc[:, :-1]

y = df.iloc[:, -1]

encoder = LabelEncoder()

y = encoder.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("\nActual Values:")

print(y_test)

print("\nPredicted Values:")
```

print(y_pred)

print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))

print("R2 Score:", r2_score(y_test, y_pred))

new_sample = [[155, 7.6, 89]]

prediction = model.predict(new_sample)

print("\nPredicted Value for New Sample:", prediction[0])

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Values")

plt.ylabel("Predicted Values")

plt.title("Linear Regression - Fruit Dataset")

plt.show()

OUTPUT:

```
   Weight  Diameter  Sweetness   Fruit
0     150       7.5         90   Apple
1     160       7.8         88   Apple
2     140       7.2         91   Apple
3     120       5.5         70  Orange
4     125       5.8         72  Orange
5     130       6.0         74  Orange
6     155       7.6         89   Apple
7     118       5.4         69  Orange
8     148       7.4         92   Apple
9     122       5.7         71  Orange

Actual Values:
[0 0 1]

Predicted Values:
[-0.20144429  0.11647736  0.74413535]

Mean Squared Error: 0.039871164528361154
R2 Score: 0.8205797596223748

Predicted Value for New Sample: 0.01251519914587984
```